



News & Social Media Analytics Engine

Data Structures and Algorithms -3
(25CS2103E)

Team Number: 02

Section: 12

Under the guidance of

Ms. K. Chandusha

Assistant Professor,
CS&IT

Submitted by:
Thota Jhana Sri
2520090121
Sutte Alekhya
2520090126

Introduction

The News & Social Media Analytics Engine is a Java-based application developed to analyze large amounts of news articles and social media posts in an organized and efficient manner. With the rapid growth of digital platforms, users receive information from many sources, making it difficult to identify which topics are popular and which information is most relevant. The proposed system processes textual data and extracts useful information such as frequent keywords, hashtags, trending topics, and sentiment patterns.

The main focus of the project is the practical application of Data Structures and Algorithms to solve a real-world analytics problem. HashMap is used for efficient frequency counting, Lists for storing and managing data, Searching and Sorting algorithms for retrieving and ranking information, and Priority Queue

Literature Survey

S. No	Existing Approach	Technology / Method	Limitations
1	News Aggregation Systems	Web APIs, Web Scraping	Mainly collect and display news
2	Keyword Analysis Systems	Hashing, Frequency Counting	Limited trend analysis
3	Sentiment Analysis Systems	Text Processing	Often focus only on sentiment
4	Trending Topic Systems	Sorting, Ranking	May not combine multiple analytics
5	Social Media Analytics	Hashtag & Keyword Analysis	Can be difficult to process large datasets
6	Proposed System	Java + DSA	Combines keyword, hashtag, sentiment and trend analysis

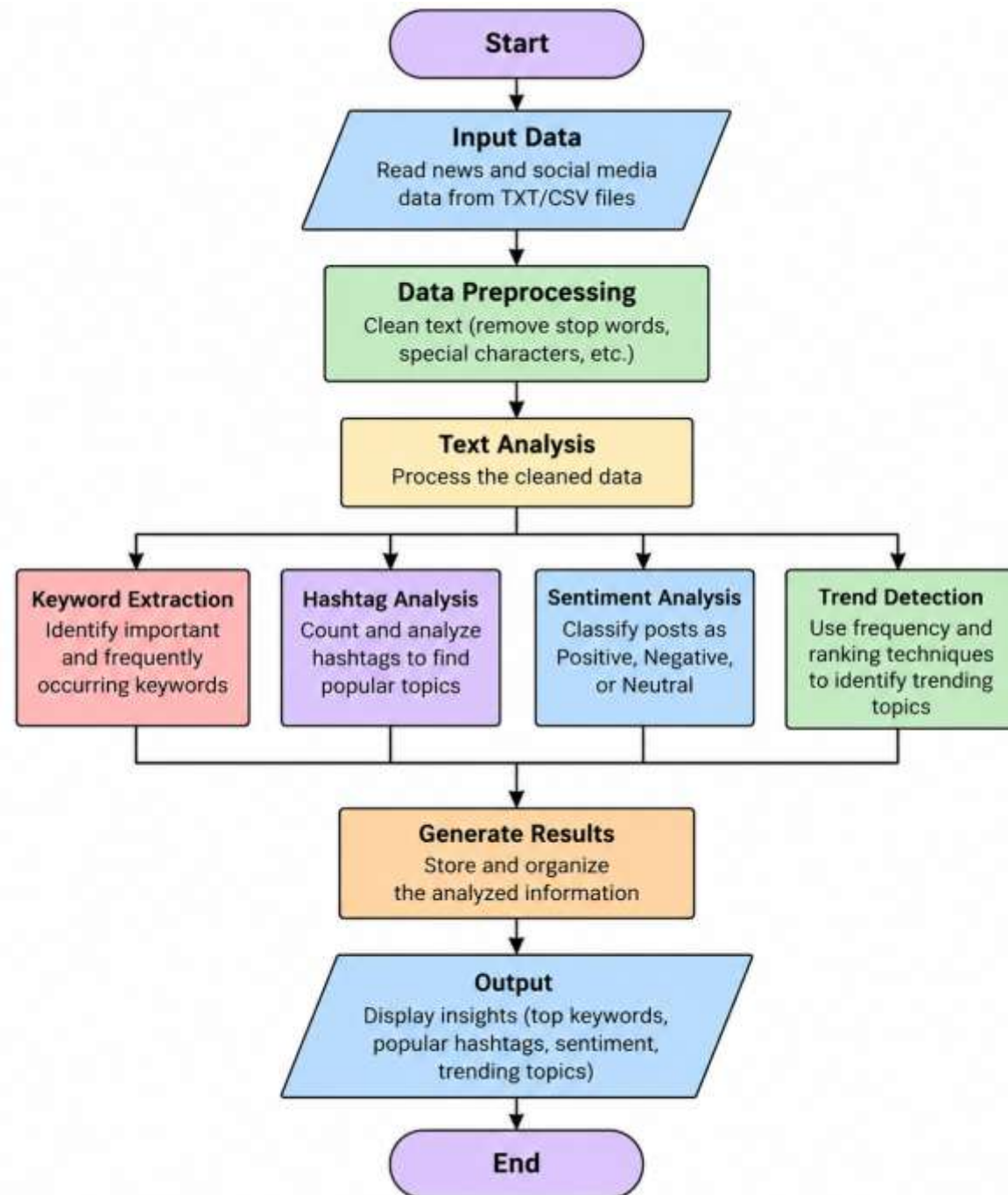
Software Requirements

Software	Purpose
Java	Main programming language
JDK 17+	Compile and run the project
IntelliJ IDEA / Eclipse	Development environment
Java Collections Framework	Implement data structures and algorithms
CSV / TXT Files	Store and read news and social media data
Git / GitHub	Version control and code management

Hardware Requirements

Component	Requirement
Processor	Intel Core i3 or above - runs the application
RAM	4 GB minimum - supports program execution
Storage	10 GB free space - stores project files and data
Monitor	14" or above - displays the application
Keyboard & Mouse	Standard - user interaction

Methodology



Conclusion

The News & Social Media Analytics Engine successfully demonstrates how Data Structures and Algorithms can be applied to analyze and organize continuously growing textual data. By processing news and social media content, the system can identify popular keywords, hashtags, trending topics, and sentiment, providing a clear overview of the information present in the dataset. The use of HashMap, Lists, Searching, Sorting, and Priority Queue makes the system efficient for storing, retrieving, counting, and ranking data. Overall, this project provides a practical and easy-to-understand application of DSA concepts while showing how efficient algorithms can transform raw data into useful and meaningful insights.

